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Artificial Intelligence in Science Education Research: Current States and Challenges

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Abstract

The recent surge of artificial intelligence (AI) in science education has heightened interest among the NARST community—a curiosity about how technology can transform education that has lasted for decades. Founded in 1928, NARST is an international organization of thousands of members focused on improving science education through research. This growing interest is evidenced by the launch of the Research in Artificial Intelligence-Involved Science Education (RAISE) Research Interest Group in 2022 and the increasing number of AI-related studies presented at NARST conferences. Despite the growth, limited studies have shed light on how the community members integrate AI into science education and the challenges. We systematically reviewed 36 AI-related papers presented at the 2024 NARST conference to address this gap. We found that AI (1) is widely integrated into science education research with ethical considerations, (2) serves to continue the paradigm of science education rather than redefining it, and (3) evolves into a multipurpose, multimodal, and generative tool for research and teaching. The analysis also revealed five major challenges: (1) changes in educational goals, (2) renovating science learning content, (3) complexity of assessment for individualized feedback, (4) teachers' preparedness and professional development, and (5) bias and equity in STEM education. Consequently, the role of NARST and RAISE RIG members that respond to those challenges is readdressed.

Keywords Artificial intelligence (AI) integration · Science education research · NARST · RAISE RIG

Introduction

The fast-growing area of artificial intelligence (AI) provides paramount opportunities to transform science teaching, learning, and assessment practices through innovating lesson planning, providing learners with personalized feedback, automating assessment, and providing unique, data-intensive experiences (Cooper, 2023; Zhai, 2023). With their collective interest in AI that traces back to the 1980 s (Good, 1984, 1987; Good et al., 1986; Stewart & Atkin, 1982; Disessa, 1987), science educators have been at the forefront in leading the integration of AI into educational research and practice (Shin & Shim, 2021; Guo et al., 2024). For example, researchers have studied automatic scoring of students' written explanations and arguments (e.g., Ha et al., 2011; Moharreri et al., 2014) or student-drawn models (Zhai et al., 2022; Lee et al., 2023) using machine learning and computer vision algorithms. More recently, science education researchers are utilizing large language models (LLMs) such as BERT, ChatGPT (Bewersdorff et al., 2023; Wulff et al., 2023), and GPT-4 V (Lee & Zhai, in press) to facilitate customized assessment and learning practices, exploring the application of AI in the preparation of pre-service teachers (Arantes, 2024), and in teaching applications more broadly (Jang and Choi, 2024). These collective efforts are upholding the AI in science education initiative

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that is pioneering the transformation of epistemic practice in science classrooms.

As AI is playing an ever-increasing role in various aspects of science education, the professional association of the science education community—National Association for Research in Science Teaching (NARST)—has recently formalized to include AI as a part of the organization's commitment, evidenced by the establishment of the Research in Artificial Intelligence-Involved Science Education (RAISE) Research Interest Group (RIG) in 2022, marking a significant milestone in the evolution of science education research that intersects with AI. Growing from an initial membership of 23, the group has quickly grown to 130 members since it was established. The 2024 NARST conference exemplified the membership's growing interest in AI with numerous presentations, including those in the RAISE RIG–sponsored poster session and Book Talk: *Uses of Artificial Intelligence for STEM Education* (Zhai & Krajcik, 2024) which attracted more than 50 participants.

Amid this situation, there is a need for evidence-based reflection on the impact of AI on science education research. As a field, we are wondering about the nature of the growing interest and applications of AI by NARST members in their work, as well as the aspects of AI being explored as research and its affordances for science education. Ultimately, the field needs forward-thinking and an understanding of the current research for future studies. To address these questions, this study reviews the presentations, posters, and workshops presented at the recent 2024 NARST conference. These outcomes advance our understanding of the status of science education research that integrates AI, identifies the challenges for the rapidly evolving landscape, and proposes future research agendas to be collaboratively answered by science educators around the globe. Using NARST's structure as a representation of science education, we sought to address the following research questions:

1. How has AI been integrated into science education research?
2. What are the characteristics of AI-integrated science education research?
3. What are the challenges that AI-integrated science education research is addressing?

Background

AI in Recent Science Education Research

The science education community has willingly integrated AI in research since the 1980s. In the early stage (1980s–2000s), AI was understood in a strong connection with cognitive science, particularly in terms of information processing theory embedded in expert systems that tried to reflect how scientists think with their knowledge base (Disessa, 1987; Good et al., 1986; Good, 1984, 1987; Stewart & Atkin, 1982). Only a few studies criticized the expert system approach, suggesting artificial neural network as an alternative (e.g., Roth, 2000). In the developing stage (2010s–early 2020s), the potential of AI in science education research rapidly evolved as machine learning methods soared in the computer science field due to the development of algorithms, hardware, and data availability (Jia et al., 2024). The major breakthrough in utilizing AI for science education revolved around automatic scoring, both in written and drawn responses (e.g., Liu et al., 2016; Anderson et al., 2018; Zhai et al., 2022), as exemplified in the special issue of *Journal of Science Education and Technology* (*JOST*) in 2021, "Applying Machine Learning in Science Assessment." The conversations between scholars with various perspectives deepened the discourse on why and how the NARST community members use AI (Li et al., 2023; Zhai & Nehm, 2023; Krist & Kubsh, 2023).

In the latest stage (2023–), the landscape of implementing AI for science education and its research has abruptly changed since the release of ChatGPT (OpenAI, 2022). While ChatGPT and other state-of-the-art generative AI models have been deemed to impact the overall education process (e.g., lesson planning, teaching material generation, formative assessment, self-directed learning, and assessment), science education researchers seem to have particularly foreseen using generative AI for research. For example, Sadler et al. (2024) pointed out that generative AI is different from previous technologies that supported research in data analysis and grammar/spelling check, as the user does not have total control over the technology with the hallucination phenomenon. Therefore, creating new knowledge and reaching openness and transparency require the researcher's commitment to critically and responsibly use AI during the research processes such as literature review, code generation during data analysis, creating outlines for manuscripts, and assisting brainstorming processes (Sadler et al., 2024). Further, a recent special issue in *JOST* also seeks "Intersections of Mind and Machine: Navigating the Nexus of Artificial Intelligence, Science Education, and the Preparation of Pre-service Teachers", in step with the "rapid advancement of generative artificial intelligence" (Cooper et al., 2025).

The studies published during this period show advanced perspectives on AI. Example studies used various AI models simultaneously as research tools. In analyzing university students' argumentation in organic chemistry, Martin et al. (2024) first used unsupervised deep learning with human interpretation to reveal argumentation patterns as clusters. Subsequently, they elicited a scoring rubric for modes of reasoning and levels of granularity in student argumentation using HDBSCAN, based on computational grounded

theory that allows complementary contributions by domain experts and machines in inductively identifying patterns in data. After human-coding the qualitative data, they tested BERT series LLMs to confirm that the deep learning model achieves almost perfect machine-human agreement (87% accuracy, 0.86 Cohen's κ, 0.86 weighted F1, and 0.80 macro F1). They further generated 20 fictitious arguments using ChatGPT to reaffirm that the deep learning model reflects human scoring guidelines in the model parameters. Ariely et al. (2024) also first conducted cluster analysis using the K-means algorithm for students' causal-mechanical explanations of the effect of smoking and anemia on physical exercise. Student responses were automatically scored using AI-grader and then classified into the previously identified clusters. They further provided students in the control group with personalized feedback based on the response classified. They reported high machine-human agreement in student response classification (87.0–90.7% accuracy, 0.880–0.933 F1, and 0.903–926 weighted Kappa) and that the control group with personalized feedback scored better than the treatment group with no feedback.

A diversity of AI-related topics are also noted. For example, Tan et al. (2024) delineated key capitals that affect students' science-related career expectations using random forest based on the Programme for International Student Assessment 2015 data. Meanwhile, other studies considered AI as the content of science teacher professional development (You et al., 2025), a shaping factor of science-related discourse in social media (Kresin et al., 2025), or which people frequently have misconceptions about what it is (Danielson et al., 2025).

In summary, since the advent of generative AI championed by ChatGPT in November 2022, the higher potential and raised concern of the evolving technology has been recognized by the NARST community. While some studies use multiple AI technologies simultaneously, exalt the status of AI, or remind AI in various research topics, the issue of ethical and responsible use of generative AI seems to be at its zenith. In this stage, it is envisaged that a deeper examination of the current status of AI integration for science education research among the NARST members could be helpful in orienting the topology of future science education research.

NARST and the RAISE RIG

This section examines how AI has become prominent in the NARST community, promoted by the RAISE RIG. In 2022, establishing the RIG named Research in Artificial Intelligence-Involved Science Education (RAISE) under NARST signaled a significant shift toward exploring the intersection of AI and science education. According to the group's charter, RAISE aims to.

> (a) support cutting-edge innovations using artificial intelligence (AI) as an emerging scientific practice to address learning, teaching, assessment, equity, and policy issues in science education; (b) improve students' understanding and practice with AI as part of being scientifically literate; (c) study the ethical dimensions of AI in its application to science education and educational research; (d) communicate the cutting-edge research involving AI to all researchers, practitioners, and policymakers; (e) encourage junior scholars in the field to pursue AI innovations within science education research as it is broadly practiced; and (f) support research in defining the boundaries and dimensions of this emerging discipline. (From RAISE Bylaws)

With this mission, RAISE focuses on increasing the quantity and quality of AI-related research within the NARST community and promoting responsible uses of AI in science education practices. RAISE's mission to advance AI in science education has led to a myriad of contributions to research, development, implementation, and policy. This community is a platform for exchanging research findings, discussing best practices, and collaborating on innovative projects. The substantial growth in membership, from 23 members in 2022 to 130 members in 2024, underscores the increasing interest in this field and the effectiveness of RAISE in bringing together a diverse group of stakeholders.

A key contribution of RAISE lies in its promotion and dissemination of cutting-edge research on AI in science education. Through annual conferences, poster sessions, and book talks, RAISE provides a forum for researchers to present their work, receive feedback, and engage in discussions with colleagues. This fosters a deeper understanding of how AI can be leveraged to address critical challenges in science education, such as enhancing student learning, developing innovative assessments, and promoting equity and inclusion. RAISE also established awards, including the Innovative Research Paper Award and the Research Worth Reading Award, to recognize outstanding research in the field. These awards not only highlight impactful research but also inspire further innovation and excellence.

RAISE is committed to promoting fairness in AI and science education. The group actively seeks to include diverse perspectives in its activities and addresses issues of bias and inequity in AI algorithms. This commitment is reflected in its membership, which includes international representation and individuals from diverse backgrounds. RAISE has also engaged with policymakers to discuss the implications of AI for science education and to advocate for policies that support the responsible and ethical use of AI in education. This has raised

awareness of AI's potential benefits and challenges in education and ensured that AI is used in ways that benefit all students.

In conclusion, RAISE has established itself as a leading voice in the field of AI in science education. Through its various activities and initiatives, RAISE has made significant contributions to research, practice, and policy. The group's commitment to building a strong community, promoting diversity and inclusion, and fostering collaboration has advanced the field and ensured that AI is used to benefit all learners.

Method

Data Collection

To investigate the current trends in using AI for science education research, this study analyzed the AI-related research presented at the 2024 NARST conference. First, the first two authors identified AI-related research included in the 2024 NARST conference by searching through the abstract and conference program books. The search terms included "artificial intelligence," "machine learning," "automatic," "automated," "computer," "chatbot," and "ChatGPT." After compiling the potential candidate reports for analysis, whether a submission (1) was related to AI and (2) was peer-reviewed (as some RIG-sponsored presentations were not peer-reviewed) was considered as the inclusion and exclusion criteria. Two researchers read the abstracts of the submissions to determine if each met the criteria. The list of analyzed documents was finalized after several rounds of discussion by the researchers, resulting in 35 abstracts. We conducted the same search for 2020–2023 NARST conference books to compare the number of AI-related studies presented each year.

We then sent emails to the authors or session organizers to request copies of the proposals for the 2024 conference. In the emails, we explicitly indicated the purpose of this review study. Twenty-four researchers responded and shared 36 proposals, including five proposals published as journal papers. This collection of papers became the data corpus for the study. The papers analyzed for this review included standalone presentations ($n = 11$), posters ($n = 7$), related paper sets ($n = 5$), symposia ($n = 5$), a pre-conference workshop ($n = 1$), and papers in the RAISE RIG–sponsored poster session ($n = 7$). Note that some abstracts in the conference books represented symposia or RIG-sponsored poster sessions, each comprising several sub-proposals. By contacting the authors, we determined the total number of presentations ($n = 50$) and collected the majority of proposals ($n = 36$) that exceeds the number of abstracts ($n = 35$) for 2024.

Analytical Framework

We devised an analytical framework to capture the characteristics of AI-integrated studies and the challenges researchers faced, including descriptive features, research characteristics, AI integration characteristics, technical features, and challenges in science teaching. Each study reviewed was assessed and coded using close- and open-ended questions for the aspects in each dimension.

The first and second authors (one Ph.D. and one doctoral student of science education) conducted a pilot analysis of 10 random-selected papers to validate the analytical framework and ensure it could capture the information from each analyzed study. The two authors discussed discrepancies in coding till consensus was met. Based on the discussion, they revised the analytical framework by adding, removing, and merging aspects in each dimension. The analytical framework was finalized after several rounds of discussion among the four researchers (Table 1).

Data Analysis

To answer research questions 1 and 2, descriptive features, research characteristics, AI integration characteristics, and technical features dimensions of the studies were analyzed. The first and second authors randomly re-sampled 10 papers and individually coded them based on the revised analytical framework. For the close-ended items, the results showed a high intercoder reliability (Cohen's kappa $= 0.837$), which is categorized as an "almost perfect" agreement according to Cohen's guidelines (McHugh, 2012). The two researchers went through several rounds of discussion to calibrate their coding. Subsequently, the two researchers each coded 13 of the remaining papers, respectively. After the two researchers had discussed several cases, the coding was finalized.

To answer research question 3, we employed qualitative thematic analysis as an inductive, subjective, and interpretative process (Terry et al., 2017) for the open-ended coding. Specifically, the coding includes five phases (cf. De Paoli, 2024): In phase 1, the first and second authors familiarized with the data. In phase 2, the two researchers assigned initial descriptive codes, took notes on each paper, and collected the evidence for the codes from the main text when necessary. For example, a quote "the introduction of ChatGPT and the following explosion in generative AI-based technologies redefine how learners and lay people can access and use science-related information." from a paper (Klein-Avraham et al., 2024) was summarized and coded as: "Learners are more and more exposed to AI-generated information."

From phases 3 to 5, we employed an AI-human collaborative approach—leveraging the ChatGPT o1-preview

Table 1 Analytical framework of the AI-related studies in 2024 NARST conference

Dimension	Aspect	Example
Descriptive features	NARST strand	1. Science learning: development of student understanding 2. Science learning: contexts, characteristics, and interactions 3. Science teaching—primary school (grades preK–6): characteristics and strategies 4. Science teaching—middle and high school (grades 5–12): characteristics and strategies 5. College science teaching and learning (grades 13–20) 6. Science learning in informal contexts 7. Pre-service science teacher education 8. In-service science teacher education 9. (Discontinued) 10. Curriculum and assessment 11. Cultural, social, and gender issues 12. Technology for teaching, learning, and research 13. History, philosophy, sociology, and nature of science 14. Environmental education and sustainability 15. Policy, reform, and program evaluation
	Education level	Preschool, elementary, secondary, higher (pre-service teachers), higher (non-pre-service teachers), higher (in-service teachers), lifelong (professional, vocational, informal), lifelong (science teacher educator, science education researcher), not specified, other
	Discipline	Physics, chemistry, life science, Earth/space science, engineering, scientific methods (e.g., observation, hypothesis), not specified, other
	Country	(Open-ended)
	Equity relevance	(T/F)
	Ethics/academic integrity relevance	(T/F)
Research characteristics	Research type	Conceptual/theoretical research, empirical research, review study, and other
	Research method	Qualitative, quantitative, mixed methods, N/A (conceptual, review), other
	Research design	Observational, quasi-experimental, N/A (conceptual, review), other
	Data collection method	Questionnaire/surveys, tests/assessments, students' artifacts, interviews, student practice observation and transcribing, learning platform, publicly open data source
	Focus of analysis	Knowledge, skills, attitudes, perceptions, engagement, practice (modeling, knowledge-in-use, competence), learning progression, N/A (conceptual, review), other
AI integration characteristics	Purpose of using AI	1. Profiling and prediction (admission decisions, course recommendation, drop-out, and retention) 2. Lesson/curriculum development assistance (material development, lesson planning) 3. Teaching competency development (creating virtual teaching settings, scenarios, recommended pedagogy) 4. Assessment and evaluation (automated grading, evaluation of student understanding/engagement/etc.) 5. Learning assistance (personalized learning, automated feedback, developing scientific understanding) 6. Other (ethics, counseling, etc.) 7. N/A
	Modality of data	Text, image, video, sound, multimodal (e.g., text + image), numeric (for prediction, clustering, etc.), N/A (no data used)
	Role of AI	1. AI is integrated into science teaching and learning activities involving research participants 2. AI is integrated as a standalone tool for research (including data analysis, developing tools/systems, etc.)
	Use of generative AI	(T/F)

Table 1 (continued)

Dimension	Aspect	Example
Technical features	AI model	(Open-ended)
	Training type	1. User-trained 2. Pre-trained/fine-tuning 3. Prompt engineering (large language model) 4. N/A (no machine learning applied) 5. Not specified (authors mentioned machine learning but did not specify its training type)
	Machine learning type (among the user-trained type)	1. Supervised 2. Unsupervised 3. Semi-supervised 4. Reinforced 5. Not specified (authors mentioned machine learning but did not specify its type)
Challenges in science teaching		(Open-ended)

model as a research tool to identify potential themes emerging from the descriptive codes (phase 3) and then reviewing the themes (phase 4) and determining the identified themes (phase 5) (similar methods refer to Kubsch et al., 2023; Sadler et al., 2024; Díaz & Delgado, 2024; De Paoli, 2024; Tai et al., 2024). OpenAI's o1-preview is recognized for its state-of-the-art higher-level cognitive inferential ability, outperforming human comparison groups in various educational domain tasks (Latif et al., 2024). The prompt used for searching themes from data is presented in Fig. 1. In phase 3, OpenAI o1-preview was once fed with the prompt and initial descriptive coding data on its chat interface, and the model identified nine candidate themes for the challenges faced in science.

In phase 4, the first author reviewed the themes identified by the o1-preview by questioning, refining, or combining them, revisiting the specific papers as needed to confirm the themes captured (Braun & Clarke, 2022; Terry et al., 2017). For example, as o1-preview suggested three assessment-related themes "challenges in multi-dimensional assessment," "large-scale data analysis and automation," and "personalization, feedback, and formative assessment," the first author combined them into "complexity of assessment for individualized feedback." In phase 5, the first author discussed the revised themes with the other three researchers and finalized five themes concerning the challenges in AI-integrated science teaching. Note that by constraining the use of generative AI and leveraging human reviews, the study confirmed the responsible uses of generative AI tools, as suggested by Sadler et al. (2024). We also kept our data from being used for OpenAI's model training to ensure data safety.

Findings

AI-Related Research Surged in the NARST Conferences

The analysis showed a consistent increase in AI-related research presentations at NARST conferences, with the number of studies increasing from 15 in 2020 to 11 in 2021, 16 in 2022, 26 in 2023, and 50 in 2024 (Fig. 2; Table 2). The number of AI-related papers presented at

Fig. 1 Prompt used to search themes using OpenAI o1-preview

> I would like to do a thematic analysis to code 36 AI-related science education research papers with the given descriptive codes/quotations for a systematic review. The item for open-ended descriptive coding was for the challenges each study tried to address about science teaching with relevance to AI. Please provide me some possible themes from the data.
>
> ---
>
> [Initial descriptive coding data]

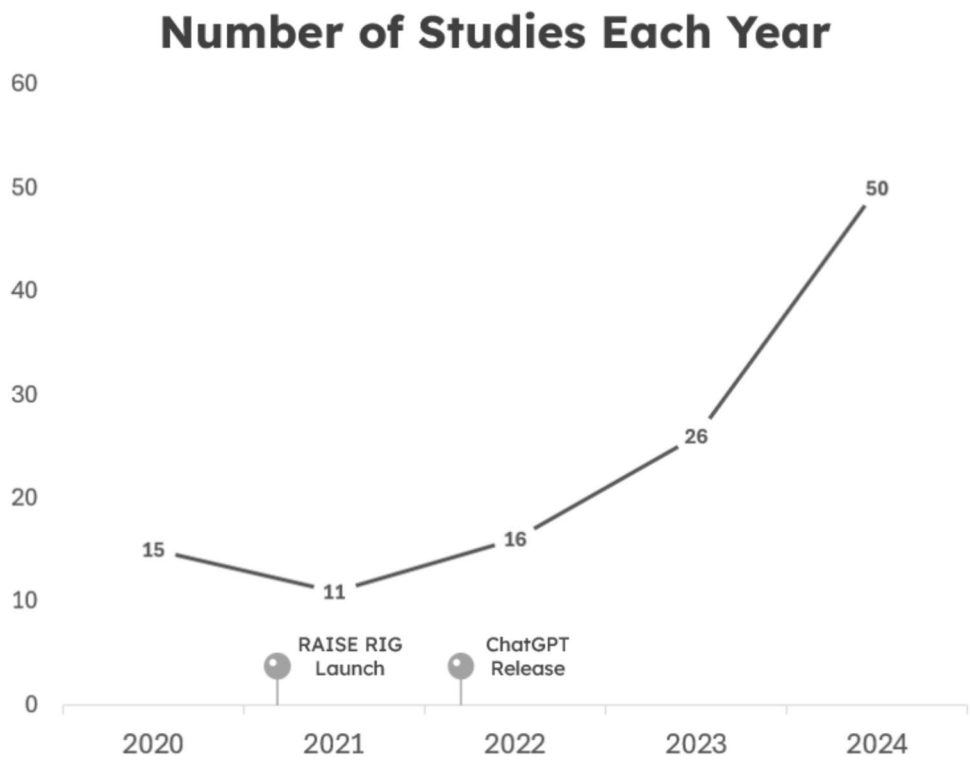


Fig. 2 Annual growth in AI-related studies presented at NARST conference from 2020 to 2024

Table 2 Number of AI-related studies presented at NARST conferences from 2020 to 2024 according to the presentation types (note: the numbers in parentheses denotes the number of sessions in conference books)

Presentation type	Year				
	2020	2021	2022	2023	2024
Pre-conference workshop					2
Stand-alone	8	6	7	10	17
Poster	1		3	1	9
Roundtable					1
Related paper set		5		2	6
Symposium	6 (1)		6 (1)	4 (1)	6 (1)
RAISE RIG sponsored session				9 (1)	8 (1)
ENG RIG sponsored session					1 (1)
Sum	15	11	16	26	50

the 2023 and 2024 NARST conferences increased significantly with the launch of the RAISE RIG in 2022, accompanying the release of ChatGPT. Note that the number of total presentations can exceed the number of abstracts in conference book, as multiple presentations can be made in symposiums and RIG-sponsored sessions (e.g., 50 presentations [among them 36 proposals were collected] from 35 abstracts in 2024; Table 2). The detailed features of the 36 AI-related studies presented for the 2024 conference are presented below.

Descriptive Features

NARST Strand

We found that most presentations were part of Strand 12—*Technology for Teaching, Learning, and Research* (18; 50%) and Strand 1—*Science Learning: Development of student understanding* (5; 14%), while a significant number of presentations were presented on the RAISE RIG (7; 19%) (Fig. 3a).

Country, Education Level, and Discipline

The majority of studies were conducted in the USA (9; 25%), Germany (5; 14%), and Taiwan (4; 11%), followed by South Korea (2; 6%) and China (1; 3%) (Fig. 3a), while many others did not specify the research context in terms of country (15; 42%). Approximately half of the studies focused on secondary education (16; 44%), followed by those targeting higher education for in-service and pre-service teachers (6; 17%). Additionally, a smaller proportion of studies also addressed higher education, targeting undergraduates who are not pre-service teachers (4; 11%) (Fig. 3c). Notably, several studies did not specify the educational level they targeted, highlighting the anticipated broad applicability of AI across various educational contexts (7; 19%). Regarding disciplinary focus, Chemistry was the most frequently addressed discipline, with seven studies (7; 19%) (Fig. 3d).

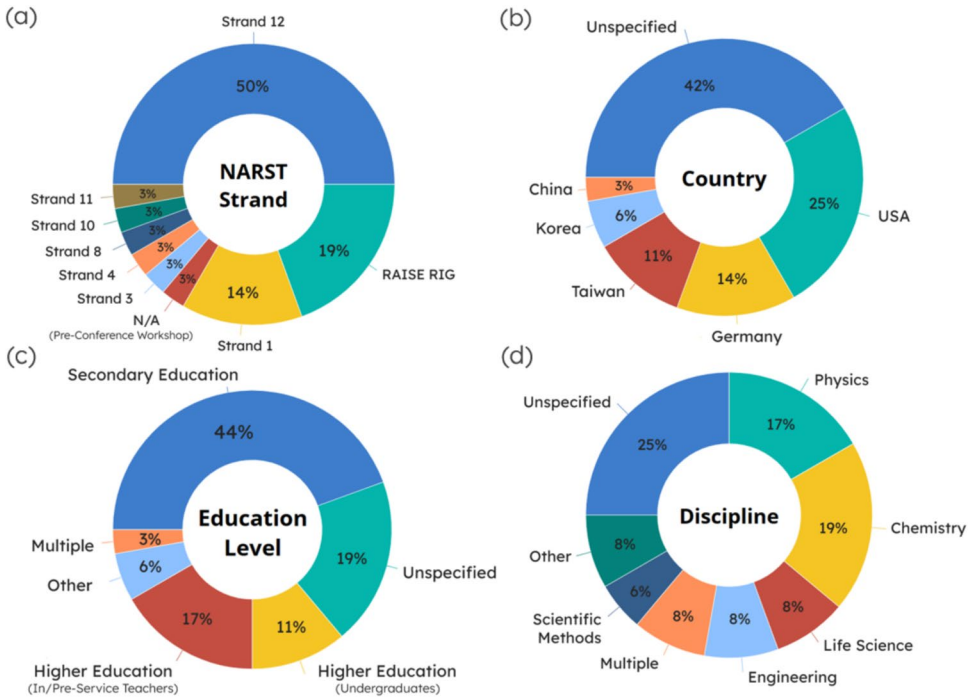


Fig. 3 Study characteristics of AI studies presented at the 2024 NARST conference

This was followed by Physics (6; 17%), Life Science (3; 8%), Engineering (3; 8%), and Scientific Methods (2; 6%). However, a notable proportion of studies (9; 25%) did not specify a target domain.

Equity, Ethics, and Academic Integrity

Markedly, one-third of the studies (12; 33%) explicitly showed each paper's focus on equity issues in using AI for science education—e.g., supporting culturally, linguistically, and ethnically diverse students with different needs in science learning (Kaldaras et al., 2024; Baranwal, 2024; Shin et al., 2024), or problematizing potential biases in AI models (Monahan et al., 2024; Kubsch et al., 2024a). Meanwhile, there were a few studies that considered the issue of ethics/academic integrity in using AI (6; 17%)—e.g., students could use ChatGPT for assignments and writing (Clark et al., 2023) and how teachers equitably respond to AI-generated feedback on students' argumentation (Mikeska et al., 2024b).

Research Characteristics

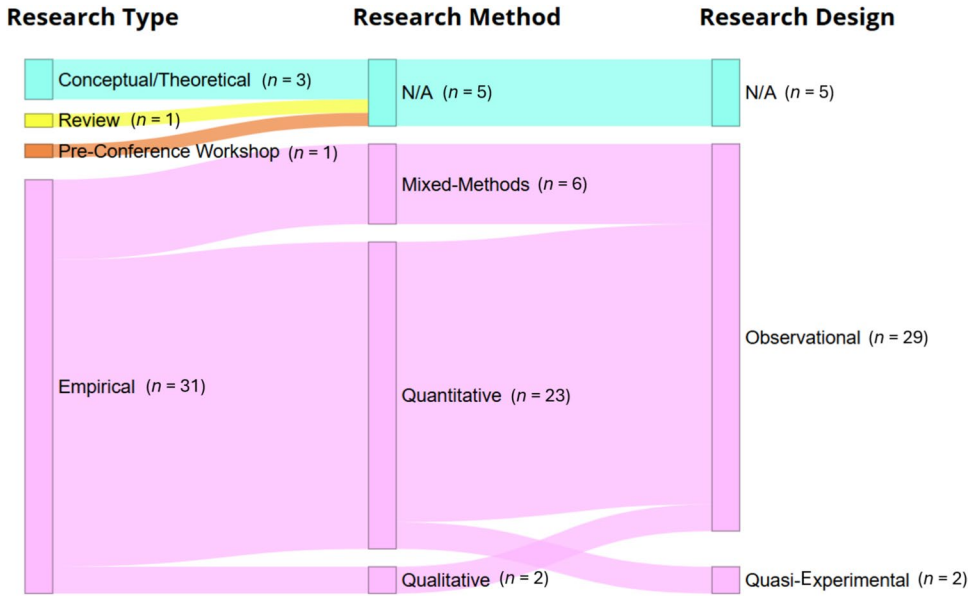
Research Type, Method, and Design

We found that most studies included empirical data (31; 86%), while only three conceptual/theoretical studies, one review study, and one pre-conference workshop were presented at the conference. Among the empirical studies, 23 (64%) employed quantitative data, followed by mixed methods (6; 17%) and qualitative data (2; 6%). However, most empirical studies were observational or non-interventional (29 of 31; 94%), while only two studies were experimental to report the effects of certain interventions (2 of 31; 6%) (see Fig. 4).

Data Collection Method and Focus of Analysis

The analysis of 31 empirical studies further identified characteristics in data collection methods and key research foci, underscoring current trends in AI-related research within science education (Fig. 5). We found that most empirical studies utilized students' artifacts as data (e.g., writing or drawing) (12; 33% of 31 studies) (Fig. 5, left). Tests/assessments (9; 25%) and questionnaires/surveys (5; 14%) were also frequently used data collection methods. On the other hand, five studies used interviews (16%), two used computerized learning platforms (6%), and one used publicly open data sources (3%).

The majority of studies focused on students' practice in science learning (8; 26% of 31 empirical studies) or learning progression (7; 23%), which are up-to-date topics in the science education research field. Among conventional constructs, knowledge (5; 16%) and perceptions (5; 16%) were dominant, followed by skills (2; 6%) and attitudes (2; 6%). Meanwhile, there were also a few studies that focused on other various constructs such as bias, science teachers' sensemaking or self-efficacy, and creativity (9; 29%) (Fig. 5, right). Note that duplicative coding was applied for the data collection method and focus of analysis.



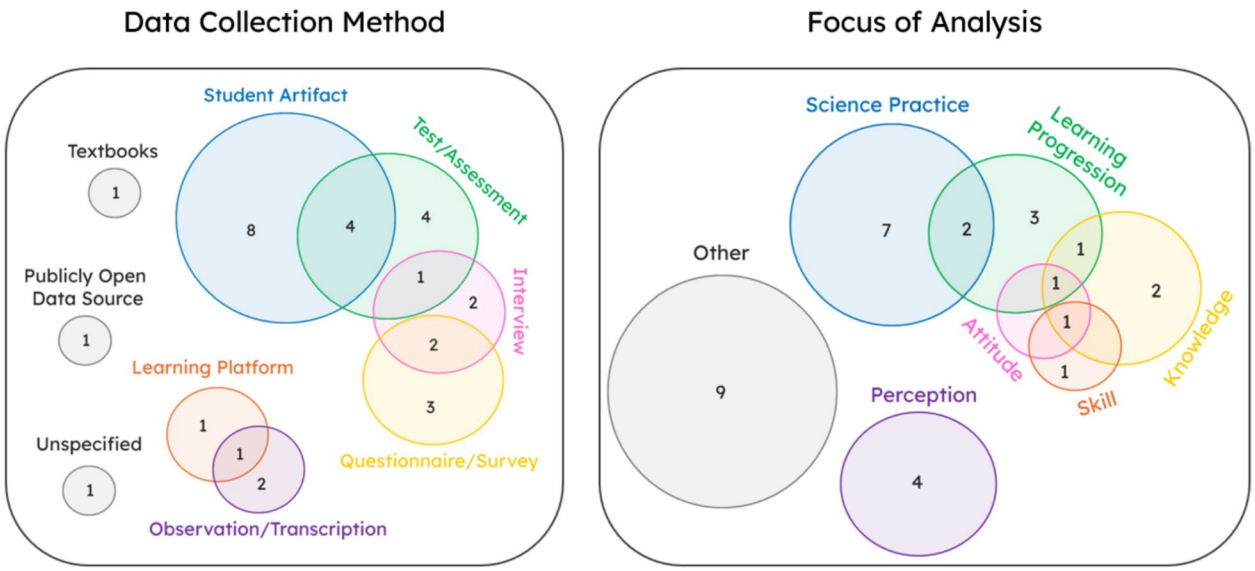


Fig. 4 Research method characteristics of AI studies presented at NARST conference

Fig. 5 Data collection method (among 31 empirical studies) and focus of analysis of AI studies presented at the 2024 NARST conference (note: the numbers in the figure indicate the number of studies)

Table 3 Characteristics of the use of AI in the studies presented at the NARST conference

Purpose of using AI	*n*	Modality of data used	*n*
Assessment and evaluation	23	Text	25
Teaching competency development	8	Numeric (for prediction, clustering, etc.)	4
profiling and prediction	5	Image	4
Learning assistance	3	Multimodal (e.g., text + image)	2
Lesson/curriculum development assistance	2		
N/A	5		

AI Integration Characteristics

Purpose of Using AI and Modality of Data

As shown in Table 3, the dominant purpose of using AI among 36 studies was "assessment and evaluation" (23; 64%), followed by "teaching competency development" (8; 22%) and "profiling and prediction" (e.g., STEM career recommendation; Yun & Crippen, 2024) (5; 14%). However, a relatively small number of studies used AI for learning assistance (3; 8%) or lesson/curriculum development assistance (2; 6%).

In terms of data modality (Table 3), the majority of studies used text data (24; 67%), followed by numeric (4; 11%) and image data (4; 11%). Two studies considered AI models processing multiple modalities in their research context (e.g., how text and image are fed to scoring model: He et al., 2024; how science teachers make sense of AI integration with various image, numeric, and sound processing models: Kim & Rachmatullah, 2024) (6%). Note that duplicative coding was applied for the purpose of using AI and the modality of data used.

Role of AI and Use of Generative AI

It is worth mentioning that most studies integrated AI as a standalone tool for research (23; 64%). While many studies used AI models to automate labeling or numerical prediction for automatic scoring, as mentioned above, some studies showed data augmentation techniques for unbalanced assessment datasets (Li et al., 2024). Another study methodologically contemplated how to validly use LLM to help the qualitative coding process (Tai et al., 2024). However, only eight studies explored integrating AI into science teaching and learning situations (22%), and all of them were related to how science teachers learn about AI or how they plan to use AI for science teaching. For example, AI was integrated into a chatbot system for simulating evolution education situations (Fiedler et al., 2024), asked to be integrated into pre-service teachers' science lesson planning (Lee & Zhai, 2024), or professional development program on AI literacy for science and non-science teachers (Moore et al., 2024). We also found that a quarter of studies employed generative AI (9; 25%), while all of them explicitly used ChatGPT (3.5 or 4).

Technical Features

AI Models Used

We found that the GPT series (including ChatGPT) (9; 25%) and BERT series (5; 14%) and other language models (e.g., bag-of-words, topic modeling, or just "NLP"—7; 19%) were the most used AI models, while conventional machine learning algorithms such as support vector machine, decision tree, and random forest were also used commonly (7; 19%). However, there were a few studies that used CNN and other computer vision models (3; 8%). Four studies utilized relatively rare AI models such as the Teachable Machine (Kim & Rachmatullah, 2024), the IBM Watson system (Shin et al., 2024a), or "a recommendation system based on the collaborative filtering algorithm, within a three-tiered design" (Shin et al., 2024b). However, eight studies did not specify or focus on particular AI models (22%). Note that duplicative coding was applied for the AI models used.

Training Type and Machine Learning Type

Most papers leveraged user-trained AI models at the researchers' end (13; 36%). On the contrary, nine papers utilized pre-trained/fine-tuning features of available LLMs (9; 25%). Seven research particularly focus on the prompt engineering of LLM (7; 19%). Seven studies did not explicitly specify how they trained machines, although they used some AI models (19%), while another six studies did not use any AI models as they were theoretical (17%).

In terms of machine learning type, supervised learning was most frequent (13; 36%), while unsupervised learning was used by five studies (14%), and four studies did not specify it (11%). There was no mention of reinforced learning or semi-supervised learning (0%). Note that duplicative coding was applied for the training type and machine learning type.

Challenges Addressed in AI-Integrated Science Education Research

We identified five major challenges by analyzing the 36 AI-based studies presented at the 2024 NARST conference. Here, we refer to Disessa's (1987) assertion regarding AI's impact on science education that "research and innovation always progress by two complementary principles, *continuity* and *opportunity*" (p. 360, original italics). While the following challenges are somehow ongoing in the science education community, they were especially outstanding with the emergence of AI.

Challenge 1: Changes in Educational Goals

We found that the reviewed studies underscored the need to revisit educational goals as AI is reshaping how we envisage the future of society and how we should prepare students for their future careers. Specifically, researchers suggest that students need AI literacy to be successful in their future lives, since learners are more and more exposed to AI-generated information. For example, "the introduction of ChatGPT and

the following explosion in generative AI-based technologies redefine how learners and lay people can access and use science-related information" (Klein-Avraham et al., 2024, p. 1). Moreover, Baranwal (2024) emphasizes the growing popularity of AI among college students and underscores and suggests that "we must communicate clearly among the learners to make them more competent" (p. 1).

Responses to the compelling AI literacy needed for students' future lives were twofold. On one side, Baranwal (2024) surveyed university students' understanding of AI, and Klein-Avraham et al. (2024) suggested that the field needs a theoretical framework for AI-based information technologies to engage students in science. After analyzing future STEM careers related to AI, such as underwater remotely operating vehicle operators, Yun & Crippen (2024) suggested that providing and maintaining the human capital for emerging and rapidly evolving careers is critical for individuals and society. On the other hand, researchers have cautioned against an overly simplistic focus on merely enhancing students' AI usage, emphasizing that addressing AI biases is integral to advancing the science literacy objectives outlined by NARST (Monahan et al., 2024).

Challenge 2: Renovating Science Learning Content

The studies pointed out the changes in what constitutes and is envisioned as science education, emphasizing the need for active engagement in redefining its scope. Specifically, they advocate for integrating AI as a core component of science education content (Kim & Rachmatullah, 2024), especially given the significant impact of AI on science enterprise. However, the effective ways to introduce its principles to students are still underexplored as "the question remains as to how to integrate AI education into core subjects … to reach all students" (Moore et al., 2024, p. 1).

Several studies addressed this issue by enlisting AI-STEM learning content. For example, using a three-round Delphi approach, Baranwal et al. (2024) identified core concepts to introduce artificial intelligence in education using robotics (AIEDuRo) at the secondary level. As a result, they listed AI introduction and AI history, Ethics of AI, Computational Thinking, Machine Learning with Scracth, etc. as the learning content of AIEDuRo. Meanwhile, Kim and Rachmatullah (2024) studied science teachers' opinions about AI experiences that students need to engage. For example, they suggested "try it out" approach for introducing various AI tools, "train it and test" approach for understanding "the aspect of AI as [science] learning agents," and "explore the epistemic" approach for deliberating "how data is used in AI and science as source of knowledge" (p. 4). However, few curriculums have been created to integrate AI into STEM learning content.

Challenge 3: Complexity of Assessment for Individualized Feedback

Studies employed AI to assess complex constructs such as knowledge-in-use and modeling that align with the three-dimensional learning of NGSS (He et al., 2024; Kaldaras et al., 2024; Li et al., 2024). Assessing such complex constructs has traditionally been a labor-intensive process for human scorers, requiring significant time and rendering it impractical for large-scale assessments—a challenge that becomes even more critical in the context of implementing formative assessments (Bleckmann & Friege, 2024). These challenges are further compounded when analyzing qualitative data, which is often plagued by issues related to validity, reliability, and reproducibility.

This challenge has also been a central focus for researchers, who have made significant advancements in the field. The integration of AI enables researchers to address tasks such as analyzing individual learning trajectories, identifying obstacles to learning, and producing individualized feedback (Kubsch et al., 2024a), as suggested in physics education (Kubsch et al., 2024b) and science-based social justice units (Wilson et al., 2024). In terms of target constructs, researchers have employed AI techniques like analyzing student-drawn models using CNN and data augmentation (Li et al., 2024) and identifying students' errors in experimentation protocols (Bewersdorff et al., 2023). Methodologically, efforts include exploring LLMs for qualitative data analysis (Tai et al., 2024), utilizing ChatGPT as an additional coder for assessing creativity (Xu et al., 2024), and advocating for computational grounded theory approaches (Tschisgale et al., 2023).

While AI models can process qualitative data with improved reliability and reproducibility (e.g., through hyperparameter tuning to control decoding process), the question of validity remains critical. The comparison between machine-coded and human-coded results is particularly important, as the validity of AI as a coder must be at least as robust as that observed among human coders.

Challenge 4: Teachers' Preparedness and Professional Development

Given the imperative to scaffold science teachers in developing their teaching competency, many reviewed studies identified the rapidly evolving educational landscape driven by AI as a significant challenge for teachers' preparedness for future classrooms. The shift is primarily attributed to the transformation of teachers' role in the classroom, as the traditional framework of "teacher as a content deliverer and students as learners in the classroom" is being redefined (Lee & Zhai, 2024, p. 1). With the incorporation of AI to support teaching and learning, the dynamic between

teachers, students, and technology is fundamentally changing (Lee & Zhai, 2024).

However, researchers have identified a significant gap in science teachers' understanding of AI, noting that many educators lack clarity on how to deliver AI-integrated instruction or familiarity with applications like ChatGPT (Kim & Rachmatullah, 2024). To address this challenge, various initiatives have been implemented to provide science teachers with experience in applying AI to science instruction and assessment. These initiatives include pre- and in-service professional development programs that equip teachers with the knowledge and skills needed to effectively integrate AI tools into their teaching practices. For instance, Fiedler et al. (2024), Shin et al. (2024a), Aydeniz and Stone (2024), and Lee & Zhai (2024) conducted research to develop teaching competencies in AI among science educators. Additionally, Mikeska et al., (2024a, 2024b) conducted studies to assess the impact of professional development programs on teachers' understanding of AI-generated feedback and their overall AI literacy. One notable contribution by Moore et al. (2024) demonstrated the effectiveness of a professional development program in enhancing teachers' interpretation of AI-generated feedback in simulated classroom settings. Similarly, Aydeniz and Stone (2024) found that their program greatly improved science teachers' AI literacy, further underscoring the importance of targeted professional development in addressing this educational challenge.

Challenge 5: Bias and Equity in STEM Education

Finally, the studies underscored concerns about bias and equity intricately related to integrating AI into STEM education. While science education researchers are eager to address these issues, these challenges have long been embedded in STEM education. Factors such as gender, ethnicity, language, and other demographic variables continue to influence student learning process and achievements in STEM, often perpetuating systematic inequities (Mikeska et al., 2024a, 2024b; Shin et al., 2024a; Aydeniz & Stone, 2024). These findings highlight the critical need for intentional efforts to mitigate biases and promote equitable practices in adopting AI within STEM education.

However, scholars have raised concerns about bias and equity issues and begun responding with theoretical and practical approaches. For example, one study emphasized the importance of differentiating the various types of "bias," such as modeling and data bias versus societal bias, to avoid confusion and position discussions toward constructive solutions (Krist & Kubsch, 2023). Another study demonstrated that algorithmic calibration helps "minimize the negative impact" of bias and "investigate the underlying source of bias" among populations participating in informal STEM summer programs (Biswas et al., 2024, p. 1). These efforts suggest that science education researchers are engaging in more profound and nuanced deliberations on bias as AI becomes increasingly integrated into classrooms, with initial solutions already being proposed to address these challenges.

Discussions

This study offers an explanation for NARST community members' interest in and utilization of AI in science education research and situates this interest in the history of this phenomenon. We first reviewed how AI-integrated studies have been reported in recent science education research literature. We then collected and analyzed 36 papers presented at the NARST 2024 conference. The characteristics of AI-integrated science education research presented at the conference were quantitatively analyzed to address the research question (1), and then the challenges the studies faced and presented were qualitatively described to address the research question (2).

AI is Widely Integrated into Science Education Research with Ethical Considerations

In terms of the number of AI-related studies and their descriptive features, we first identified the pattern of the soaring number of AI-related studies presented at the NARST conference from 2020 to 2024 (from 11 to 35, according to the conference books). The consistent upward trend highlights a deepening interest in AI within science education and underscores RAISE's role in catalyzing this area of research. Also, the significant growth in AI studies demonstrates the expanding recognition of AI's potential to enhance educational methodologies and outcomes, as exemplified in NARST strands, education levels, disciplines, and countries.

While the prevalence of Strand 12: *Technology for Teaching, Learning, and Research* seems natural, considering AI as cutting-edge technology, the variety of strands presenting this work supports the scalability of AI-integrated science teaching and the evolved feasibility of the technology. If AI becomes more popular in science classrooms, AI could be interpreted and adopted in more diverse learning situations reflected in the NARST strands beyond just a technology tool (Díaz & Delgado, 2024). Particularly, we highlight the role of the RAISE group as it contributes a significant portion of AI-related studies at the NARST conference, which is deemed to continue in the future.

Science education researchers may have considered the secondary level as the most suitable and crucial educational site to integrate AI. In contrast, the scarcity of research in the primary education context may indicate the scholars' concern for the safe use of AI for younger learners (e.g., under age 13—UNESCO, 2023). School districts' active and

apprehensive role in providing access to this form of technology may also be a consideration (Crompton et al., 2022; Danjoux et al., 2024; Wang & Cheng, 2021). The inclusion of pre-/in-service teachers besides higher education learners implies science education researchers' interest in equipping science teachers with AI technologies for the future, which will be discussed further below.

The balanced disciplines and countries as research contexts of AI-integrated studies reassert that AI integration is a global trend in science teaching research that is considered more and more important. Whereas AI seems to have served conventional subject domains of science such as physics, chemistry, and life science, how AI itself as an interdisciplinary field could serve more interdisciplinary STEM courses worth being explored (Cai et al., 2024). On the other hand, the studies were led by both Western countries (e.g., USA and Germany) and Eastern countries (Taiwan, Korea, and China). This evidence supports the diverse perspectives raised and shared at the NARST conference and the international focus of the organization.

The AI-related presentations at the 2024 NARST conference focused on equity, ethics, and academic integrity, assuring that the NARST community members' interest did not only pertain to the mere application of the technology but also to the good use of it. This echoes the globally emerging discourse on the responsible use of AI for education (UNESCO, 2023; Nguyen et al., 2023), showing science education researchers' timely reflection and response to the concerns.

In summary, the NARST community is showing a growing, diverse, and thoughtful standpoint in integrating AI for science research, not unquestioningly adopting fancy technology. In addition to the oughtness to pursue diversity and mitigate potential biases in AI models and data, we recall that one audience in the AI-related session at the 2024 NARST conference raised a question about data confidentiality, asking whether it is ethical to upload student data onto commercialized AI models such as ChatGPT or Gemini to analyze it. This is particularly important in terms of potential breach of FERPA (Family Educational Rights and Privacy Act) or IRB (institutional review board) guidelines. Although we can nowadays set commercial AI services not to use our data for model training, using open-source AI models that operate locally in the researcher's computing system can be an answer to those questions to further enhance the ethical use of AI for science education research.

AI Serves to Continue the Paradigm of Science Education Rather Than Redefining It

Regarding research characteristics, we primarily discovered that most studies were empirical in nature, while several were conceptual/theoretical. This shows science education researchers' positivistic attitude in examining the educational changes brought by AI into science classrooms. However, the prevalence of quantitative methods and observational research design implies that there is still room for deeper qualitative investigations on the phenomena or quasi-experimental scrutinization of the impact of technology on student learning outcomes.

The utilization of student artifacts and test/assessment data, as well as interviews and questionnaires/surveys, indicates that science education scholars are employing traditional types of research data for AI-related studies over already digitized data gathered from computerized learning platforms. This strongly evidences science education researchers' orientations toward realistic classroom situations rather than exceptionally technology-enhanced learning environments.

Also, the focus on science practice and learning progression, in addition to conventional knowledge, skill, attitudes, and perception, shows the complex constructs contemporary science education researchers try to improve in science learners. Notably, the focus of analysis aligns with the concurrent paradigm of science education, such as NGSS (NGSS Lead States, 2013) and learning progression theory, not going unoriented with the tide of general trends for research on AI in education (AIEd).

Synthetically, the patterns from the number of AI-related studies to focus of analysis indicate that the NARST members are employing AI to facilitate the science education the community has imagined and shaped rather than deconstructing and reconstructing what science education means and does. For example, the Next Generation Science Assessment (NGSA) initiative aims to "designing formative assessments for classroom use aligned with the Next Generation Science Standards [three-dimensional learning]" considering AI's affordance, rather than redefining to goals of science assessment (NGSA, n.d.; see also Zhai et al., 2020). The AI-related science education presentations at the 2024 NARST conference need to be understood in a similar keynote to stand for the current science education paradigm of the timepoint, which will also be the case for the future.

AI Evolves into Multipurpose, Multimodal, and Generative Tools for Research and Teaching

In terms of AI integration characteristics and technical features, we first identified that automation of assessment and evaluation was the primary purpose of using AI in science education. Also, most studies used data in text modality. However, we stress that studies integrating AI are leaping over the wall, serving other purposes such as teaching competency development, profiling, and prediction. On the other hand, it is encouraging that several studies utilized image data or focused on AI models' capabilities of multiple modalities beyond text. As general AIEd discourse projects

that AI could exploited for essentially every point of educational practices (from planning to facilitating teaching, learning, and assessment) with multimodal reasoning ability (Lee & Zhai, in press), we could easily expect that would also be the case for science education (e.g., analyzing lab safety rules compliance with the given photo of chemistry lab; Lee & Zhai, in press).

Science education researchers' usage of AI as a research tool seems to be due to its popularity for automatic assessment, which is quite established, and its potential to be used for qualitative coding, which is recently emerging (as in this study). However, the role of AI needs not remain as a research tool, as several studies already pointed out that AI could be introduced to science teaching and learning situations. Therefore, a remaining agenda in AI integration for science education is actively engaging teachers and students in interacting or collaborating with AI in classrooms (Jeon & Lee, 2023; Tang & Cooper, 2024).

Remarkably, GPT, BERT, and other LLMs played significant roles in the AI-related studies presented at the 2024 NARST conference, compared to traditional machine learning algorithms such as decision trees and support vector machines. We interpret this in two dimensions: (1) Highly user-friendly interfaces of those new AI models facilitate their integration by researchers, mitigating previous access barriers. (2) Generative AI models that return natural language responses tailored by prompt engineering make it easier for them to be employed as educational agents for various purposes, compared to the traditional models that merely yield statistics. However, as there is an atmosphere of studying AI technologies for science education research among the NARST members, more advanced techniques such as fine-tuning, CNN, semi-supervised learning, and reinforced learning are also projected to be utilized in the upcoming years.

Consequently, we posit that AI integration in the science education research field will head toward the multipurpose, multimodal, and generative tool for research and teaching tool. To satisfy science educators' future needs for this, large multimodal models (LMMs) such as OpenAI's GPT-4o and Sora and Google's Gemini 1.5 Pro that process and generate text, image, audio, and video would be eagerly imported. This will eventually help science education practitioners realize the principles of multimodal learning and assessment in usual science classrooms.

Challenges Revealed in AI-Integrated Science Education Research and Future Directions

The 2024 NARST conference presentations related to AI reveal the challenges contemporary science education is facing by revisiting and tackling them thanks to technology (Disessa, 1987). The future directions of AI-integrated science education corresponding to the challenges identified, which science education researchers or practitioners should put endeavor, are as follows:

(1) *Changes in educational goals*: The 2024 NARST presentations' notice of students facing AI-generated knowledge matches with the current discourse on "AI [that] is transforming how science is done" (Erduran, 2023). Therefore, we must contemplate how to devise the steppingstones (i.e., developing AI literacy) and where to set our goals (preparing students for future AI-related STEM careers). However, we insist that AI integration is not to seek something beyond or other than science education but to awaken science educators to be mindful of the current science teaching practices, as scientific literacy can be expanded to embrace AI literacy (Cheung et al., 2024). To this end, we could use literature from the machine learning or statistics field that intersects the broader meaning of scientific "learning" from the environment (Box, 1979; Newell, 1982; James et al., 2023) to further our understanding of scientific learning.

(2) *Renovating science learning content*: Adding on to the 2024 NARST presentations' acknowledgment of AI as a potential STEM learning content, the 2024 Nobel Prizes for Physics and Chemistry were both related to AI and will be another driving force for this renewal as they signify the impact of AI on our society. Thus, it is likely to be an innovation in K-12 science education to embrace the principles and applications of AI as learning content. While there was a lack of curriculum-aligned studies, we project the AI-led transformation to improve how we enact science education in schools so that we cohesively pursue the current paradigm of science education—e.g., NGSS-aligned science education in the case of the USA. Thus, the future directions of AI-integrated science education research will start from the synthetic, comprehensive, and overarching framework for integrating AI as the content of STEM education.

(3) *Complexity of assessment for individualized feedback*: The multifaceted constructs assessed with AI reassert the previously unforeseen affordance given by AI to pursue the NGSS-aligned learning goals. As three-dimensional learning is challenging to assess and needs tremendous labor, introducing AI for automatic assessment is again supported. Further, real-time individualized feedback to science learners was not feasible in a typical science classroom teacher-many-students structure but is one of the most rapidly progressing AIEd research fields (Maier & Klotz, 2022). We can understand this formative assessment system in two steps: (a) assessing students' learning status accurately and

fast and (b) generating and providing feedback to students based on the assessment rubric. Previous studies seem to have focused on the former to build reliable automatic scoring systems; however, the latter seems underexplored and thus needs follow-up research.

(4) *Teachers' preparedness and professional development*: While the science education research community is progressing in terms of utilizing AI, equipping pre- and in-service science teachers with AI-integrated teaching competency is a remaining task. Providing pre-service teachers with introductive AI courses and inviting in-service teachers with AI professional development programs are strongly requested. Particularly, as generative AI models can work as educational agents based on knowledge retrieval and natural language–based communication, the changing roles of science teachers—at least, departing from the role of the sole educational agent in the classroom—should be addressed to reposition themselves.

(5) *Bias and equity in STEM education*: Reducing biases in every step from planning to enactment has been of major importance in shaping more equitable STEM education, echoing fairness that has been seriously considered in science education (Guzman-Orth et al., 2021). We appraise that the scholars have already been wrestling with bias and equity in AI integration, not only theoretically but also technically. However, we raise the following points for future studies. First, mitigating bias does not directly lead to ensuring equity. To analogize, the former is like curing disease from a person (from negative to neutral states), and the latter is like establishing healthier and lively well-being (from neutral to positive states). Second, AI integration could be harmful or helpful in pursuing our sublime ideas in science education, depending on how successfully we harness the technology for our lives. Thus, the NARST community members' endeavors are needed to significantly enhance ethical and responsible training, testing, use, and management of AI models for science education. Guidelines published by international organizations such as UNESCO (2021; 2023), OECD (2023), and EU (European Commission: Directorate-General for Education, Youth, Sport and Culture, 2022) can be helpful in pursuing this goal.

Conclusion

OECD (2019) characterizes the current era as uncertain due to the fast-evolving AI and stresses that education should not fall behind the development of technology. Rapidly evolving AI technologies are already racing the track, and innovative AI applications in science education research should and will chase them to transform the landscape of science education. This prompts self-reflection and mindfulness of our actions, purposes, roles, and practices in science education. If AI had not fundamentally reshaped science education as of 2020 (Zhai et al., 2020), can we still assert this as of 2024? Good et al. (1986) said "there were no dramatic changes in science education [due to AI] in the year 1984; perhaps 2001 will be different" (p. 273). While there were also no dramatic changes in the opening of the twenty-first century, we have been facing such a dramatic change since 2022. Although the obvious change has been delayed about in the last 20 years, the impact is never smaller than our predecessors' expectations, as AI is agitating today's science education globally. However, we could again cite Good et al. (1986) in that "human science teachers and science education researchers will not be replaced by the "intrusion" of expert systems [and any other AI models] into education, but surely their roles will change" (p. 273). Indeed, the future of science education and research is being reshaped in step with the tide of AI, but by us—the science education community members of NARST and RAISE RIG, who collectively, enthusiastically, and humbly respond to those challenges.

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